📘 COMPREHENSIVE COURSE CURRICULUM

Deep Learning Specialization

Module 1 Artificial Neural Networks (ANNs)

#	Topic	What You'll Learn
1	**Deep Learning Basics & Intuition**	Why DL beats traditional ML — powered by big data & hardware (GPUs, TPUs).
2	**Perceptrons**	The conceptual building blocks of neural networks.
3	**Forward Propagation**	Step-by-step flow of data through the network.
4	**Backward Propagation**	The learning process, including the weight update formula.
5	**Vanishing Gradient Problem**	Why certain activation functions hinder deep network training.
6	**Activation Functions**	Deep dive into ReLU, Leaky ReLU, PReLU, and Swish.
7	**Model Implementation Basics**	Building a model using TensorFlow and Keras.
8	**Loss Functions**	How models quantify error — Binary Cross-Entropy, Mean Squared Error.
9	**Optimizers**	Training techniques — Gradient Descent, SGD, Mini-Batch.
10	**ANN Project**	Hands-on: building a complete ANN model.
11	**Black Box vs. White Box Models**	Discussion on model interpretability.

Module 2 Convolutional Neural Networks (CNNs)

#	Topic	What You'll Learn
1	**CNN Basics & Intuition**	Why CNNs are essential for image data — spatial structure, parameter efficiency, and translation invariance.
2	**Convolution Operation**	How filters/kernels slide across input data to detect features — edges, textures, patterns.
3	**Padding & Stride**	Controlling output dimensions — valid vs. same convolution, and how stride affects spatial size.
4	**Pooling Layers**	Downsampling techniques — Max Pooling and Average Pooling for dimensionality reduction.
5	**Fully Connected Layers**	How extracted features are flattened and fed into dense layers for final classification.
6	**CNN Architecture Overview**	The full pipeline: Conv → ReLU → Pool → Flatten → FC → Output.
7	**Training CNNs**	Backpropagation through conv layers, gradient descent, and the weight update process.
8	**Regularization Techniques**	Dropout, Batch Normalization, and L2 Regularization to prevent overfitting.
9	**Data Augmentation**	Artificially expanding training data via rotation, flipping, scaling, and noise injection.
10	**Landmark Architectures**	Evolution of CNNs — LeNet, AlexNet, VGG, ResNet, and their key innovations.
11	**Residual Connections (ResNet)**	Shortcut connections that solve the vanishing gradient problem in deep networks.
12	**Transfer Learning**	Using pre-trained models like VGG and ResNet for new tasks with limited data.
13	**CNN Project**	Hands-on: building a complete CNN for image classification (e.g., CIFAR-10, Fashion MNIST).

Module 3 Sequence Modeling (RNN, LSTM & GRU)

#	Topic	What You'll Learn
1	**Foundations of Sequence Modeling**	Why standard ANNs and CNNs fall short when dealing with sequential and time-based data.
2	**RNN Architecture**	Deep dive into internal mechanics — hidden states and time-steps for memory maintenance.
3	**Technical Challenges**	Vanishing and exploding gradient problems during training of traditional RNNs.
4	**LSTM (Long Short-Term Memory)**	Memory cells, gates (forget, input, output), and how they improve long-term dependency learning.
5	**GRU (Gated Recurrent Units)**	Alternative architecture — simplified gate structure compared to LSTM.
6	**Next Word Prediction Project**	Practical implementation using TensorFlow and Keras — data preprocessing, tokenization, model training, and deployment.

Course Overview & Details

Prerequisites

- Strong foundation in Python programming.
- Basic understanding of Linear Algebra and Calculus.
- Familiarity with Pandas and NumPy libraries.

Target Audience

- Data Scientists expanding into Deep Learning.
- Software Engineers transitioning to AI/ML.
- Students pursuing advanced AI research.

Tools & Frameworks

- **Primary Frameworks:** TensorFlow 2.x, Keras
- **Data Handling:** NumPy, Pandas
- **Environment:** Jupyter Notebooks, Google Colab

Outcomes

- Ability to architect custom neural networks.
- Portfolio of 3 end-to-end Deep Learning projects.
- Readiness for advanced topics (Transformers, GANs).